

Harvesting Power at High Frequencies: Building an RF Power Combiner and Rectarray as a Sustainable Alternative to the Traditional Battery

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Background

- Traditional batteries harm the environment
 - Contain hazardous materials (Pb, Cd, Hg)
 - Improper disposal can lead to soil and water pollution
 - Regular replacement contributes to electronic waste
 - Extracting natural resources has environmental and social concerns



According to the Environmental Protection Agency (EPA), only about **5% of lithium-ion batteries** in the US were recycled in 2022

Energy Harvesting: The conversion of ambient energy present in the environment into electrical energy for use in powering autonomous electronic devices or circuits

- Harnessing RF energy around us to power devices
 - Ultra-High Frequencies from cellular towers

Research question

What is the peak power output achieved by a harvester board design with seven rectifiers when receiving signals from a UHF source?

Hypothesis

If a harvester board is designed with a topology including 7 rectifiers, then it would produce a power output but less than the source signal's <1 W of power output because of the additional rectifiers enhancing the harvesting capability. Assuming a 5% efficiency, this would result in a power output of 50 mW.

Methodology

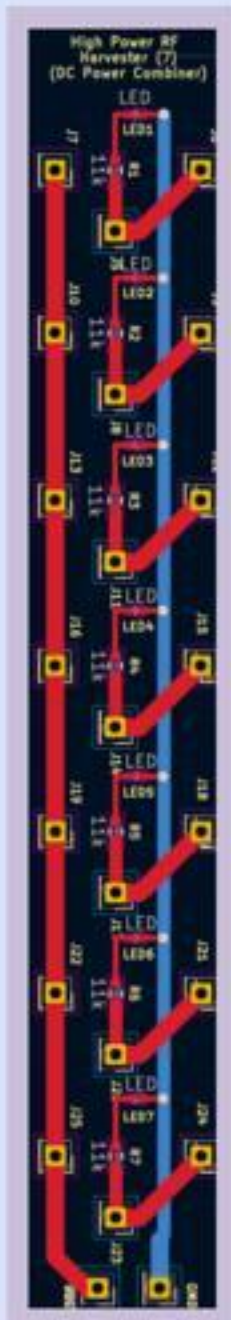
- Design and build a rectifier array Harvester Board and Power Combiner with 7 rectifiers
- Test prototype with a 749 MHz signal to produce a DC output voltage
- Identify peak power output and analyze experiment results

Design

KiCAD



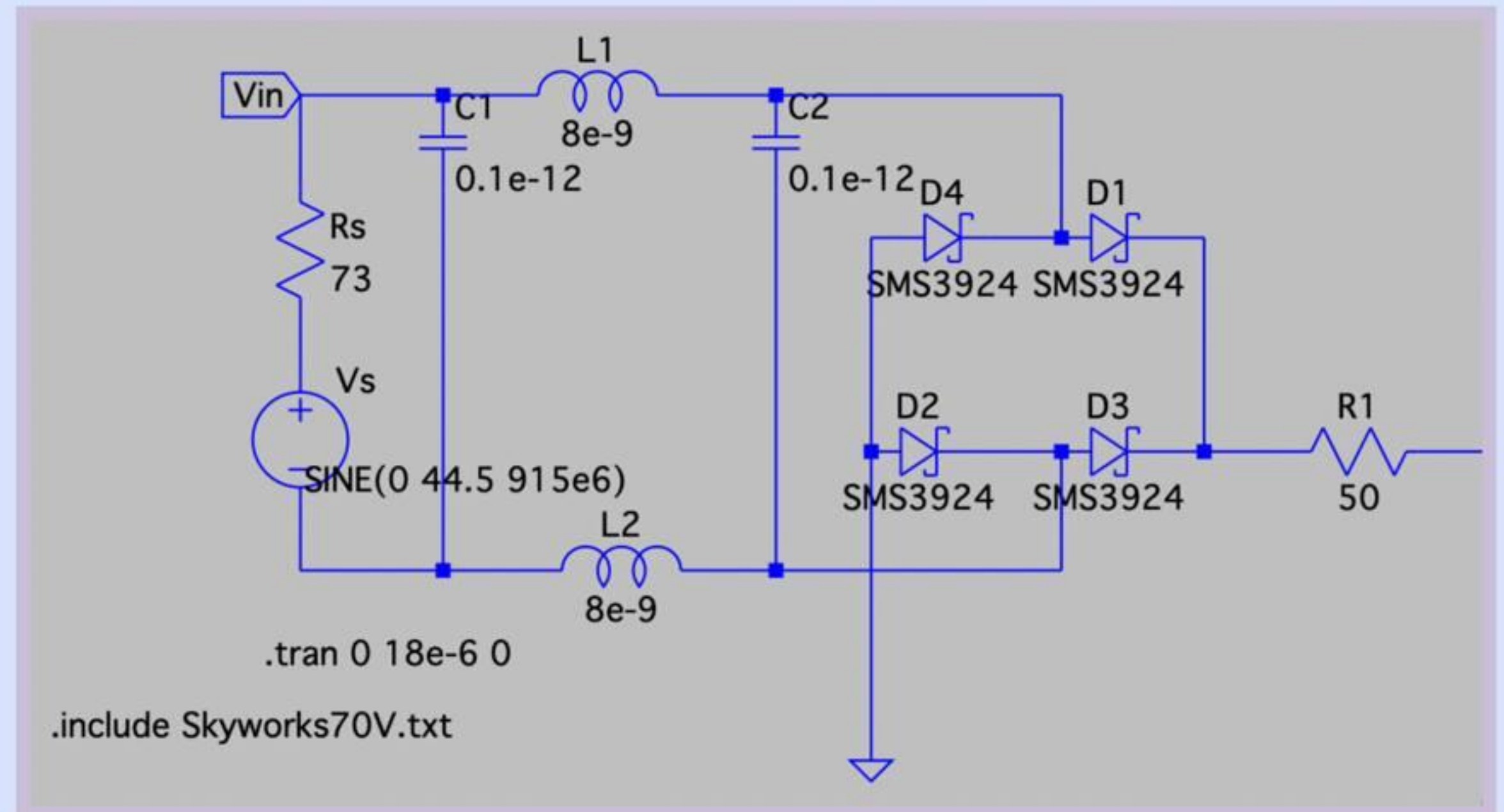
Rectarray



Power combiner

LTSpice

Diagram of one
rectifier

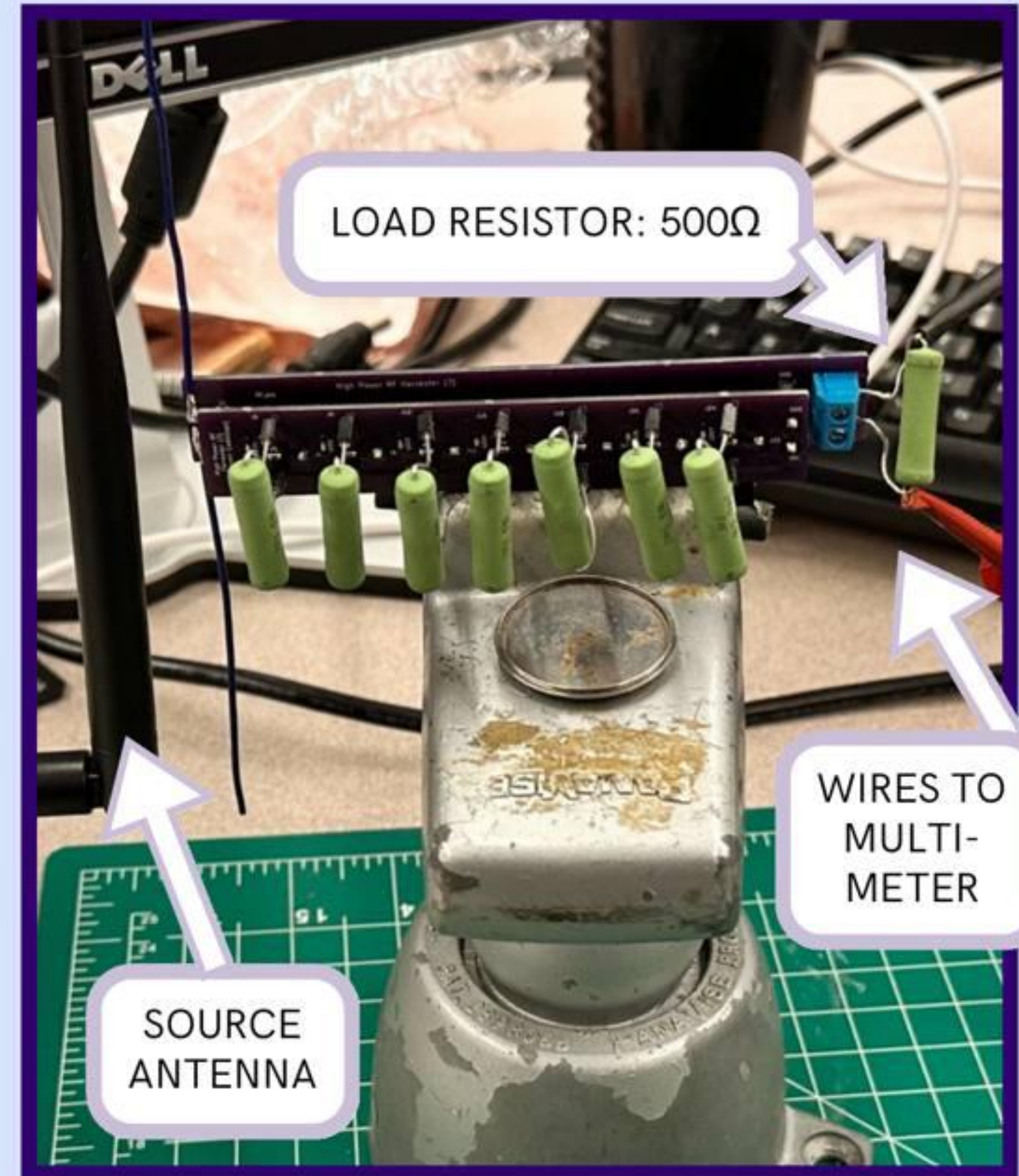
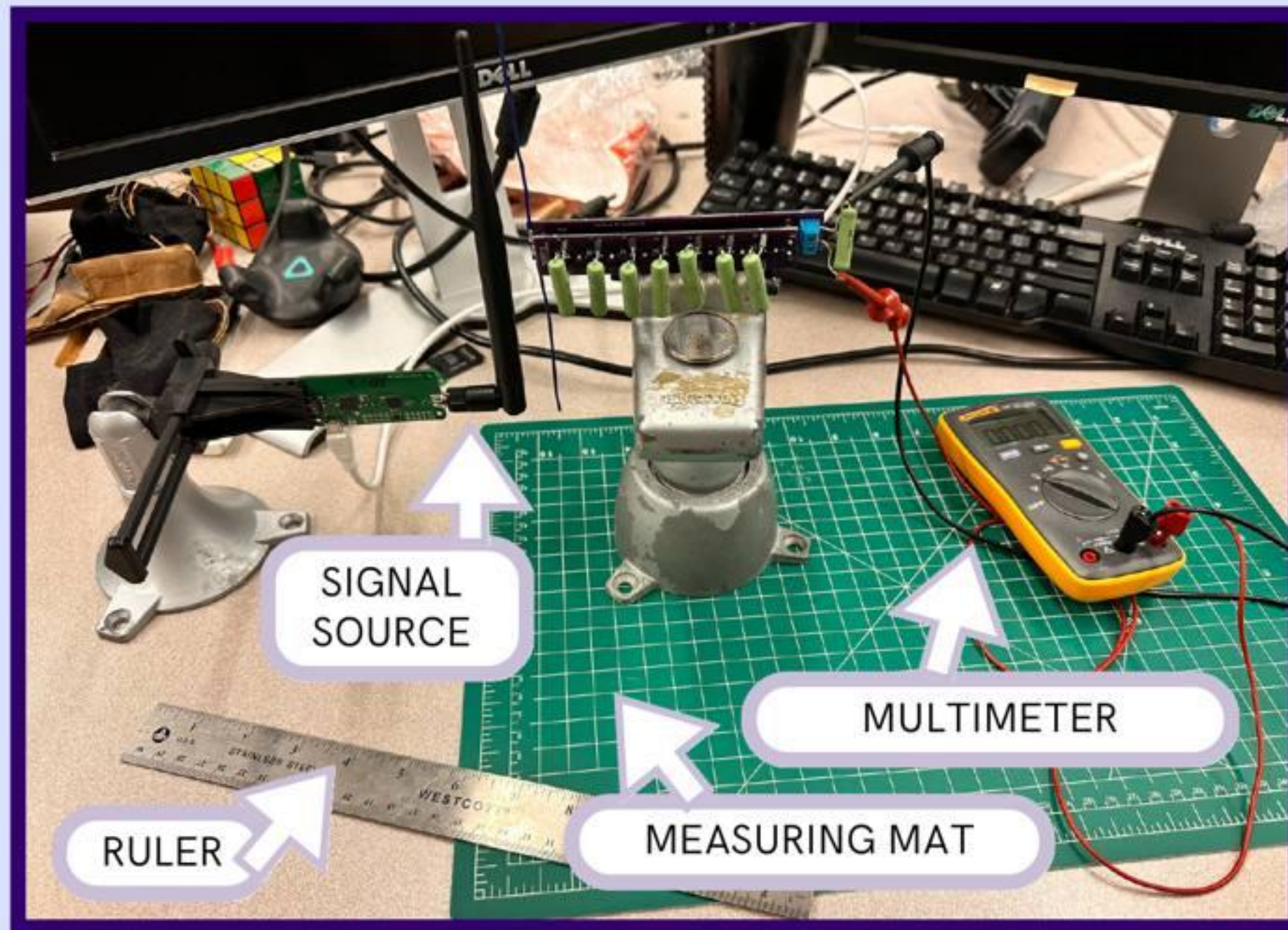


Building prototype



Data analysis

Experimental Set-up

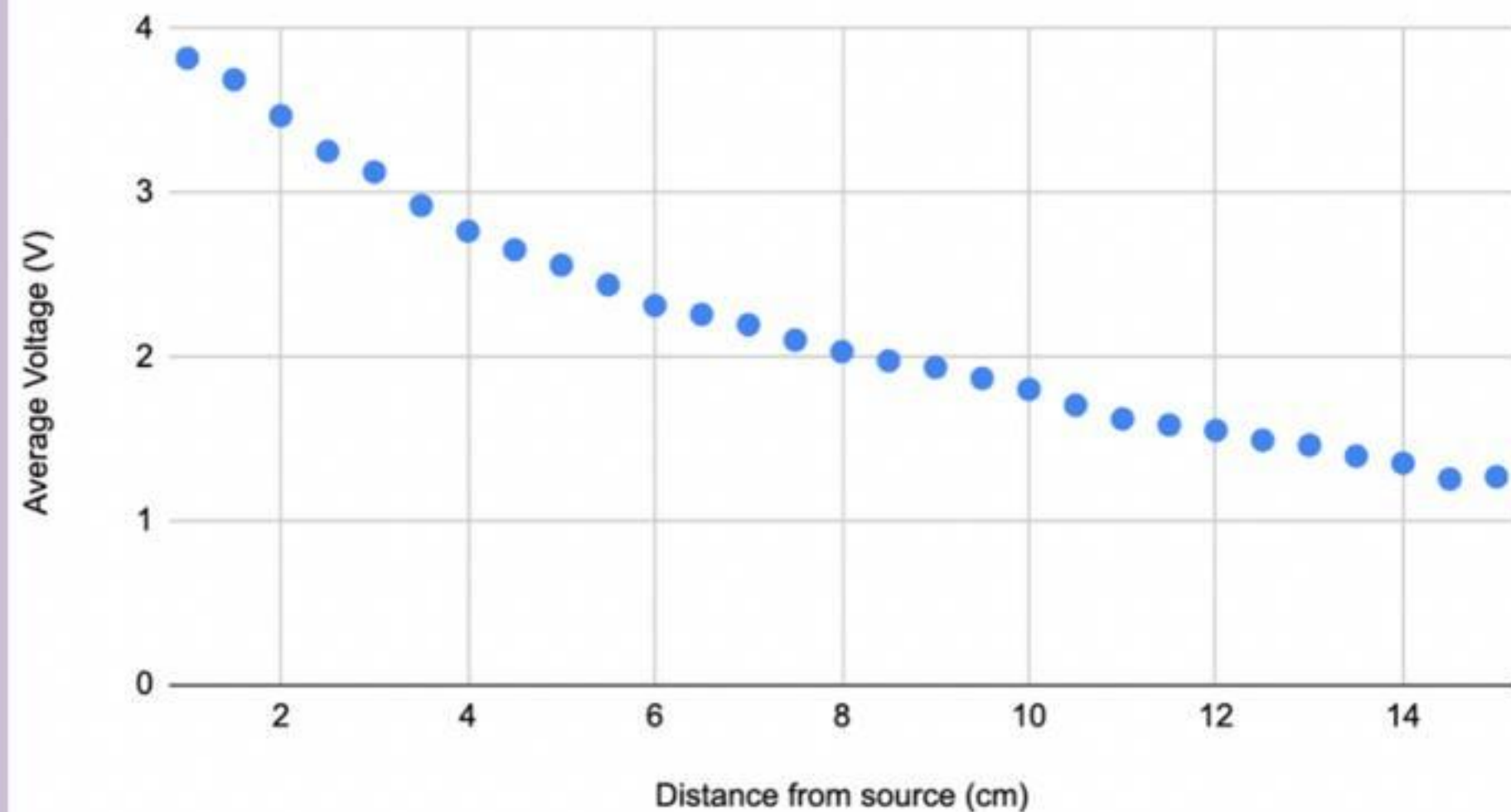


Results

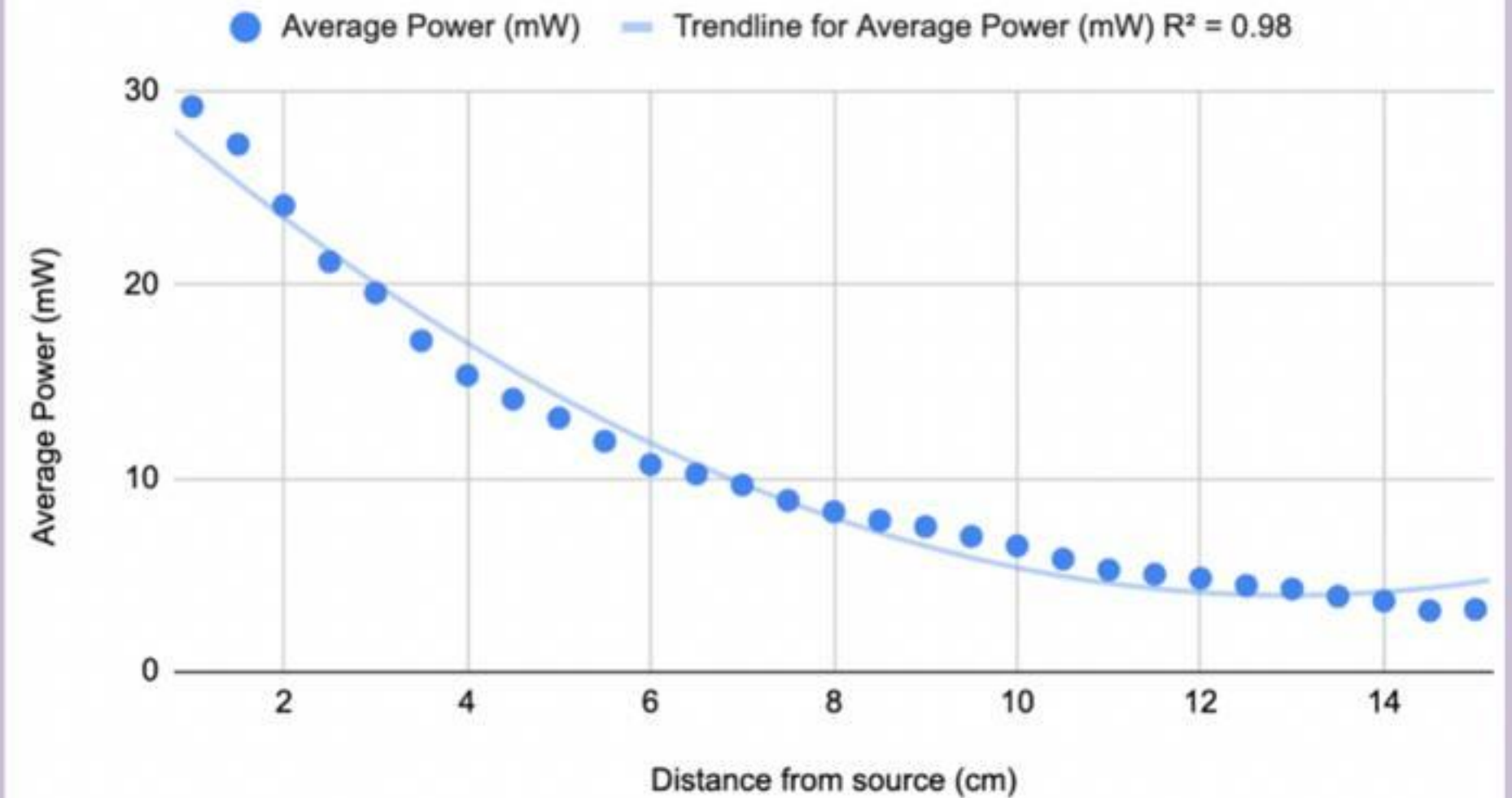
Peak power output: 29.18 mW

~3% efficient

Average Voltage (V) vs. Distance from source (cm)



Average Power (mW) vs. Distance from source (cm)



Conclusions

In this research, an overview of the design, implementation, and test results of a UHF harvester rectarray prototype and power combiner are presented, and future applications and potential research are introduced. A peak power output of 29.18 mW is achieved from this prototype, boasting a 3% efficiency from its 1W UHF signal source. This research shows promising results for the future of energy harvesting, with potential for even greater results from a larger signal source, and for sustainable solutions to traditional energy sources.

Future work

Next steps:

- Explore different designs for the harvester
- Use a different antenna design
- Wider range of compatible frequencies
- *Use a high-power input source to generate more power output*

References

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